

NyotaPay Payment Platform

Enterprise Architecture Document v2.0

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Executive Summary

NyotaPay is a comprehensive payment platform designed for on-premises deployment, supporting P2P transfers, cash-in/out operations, and merchant payments across multiple channels (Mobile, USSD, SMS). This document presents an improved architecture leveraging modern design patterns, cloud-native principles, and financial services best practices.

Key Improvements

- **Microservices with Domain-Driven Design (DDD):** Clear bounded contexts and service boundaries
 - **Event-Driven Architecture:** CQRS and Event Sourcing for scalability and auditability
 - **Enhanced Security:** Zero-trust architecture, service mesh, and advanced threat protection
 - **Cloud-Native Patterns:** Container orchestration, service mesh, and infrastructure as code
 - **Improved Resilience:** Circuit breakers, bulkheads, retry policies, and chaos engineering
 - **Advanced Observability:** Distributed tracing, metrics, and centralized logging
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Table of Contents

- 1. [Architecture Principles](#)
- 2. [System Context](#)
- 3. [Architecture Patterns](#)
- 4. [Component Architecture](#)
- 5. [Security Architecture](#)

6. [Data Architecture](#)
 7. [Integration Architecture](#)
 8. [Deployment Architecture](#)
 9. [Resilience & Scalability](#)
 10. [API Design](#)
 11. [Observability](#)
 12. [Disaster Recovery](#)
-

1. Architecture Principles

1.1 Core Principles

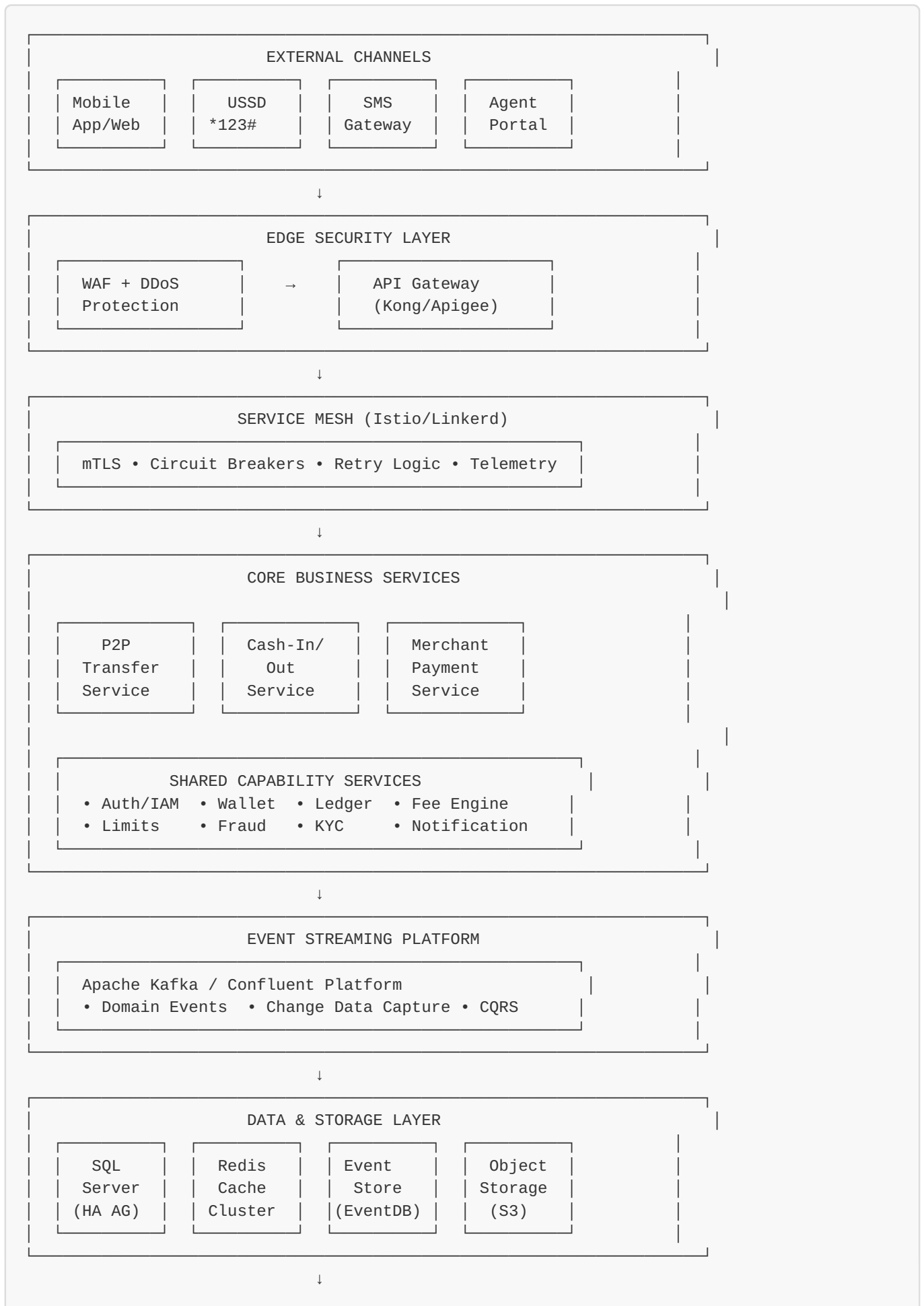
1. **Security First:** Every component implements defense-in-depth
2. **Financial Integrity:** Strong consistency for money movements, eventual consistency elsewhere
3. **Scalability by Design:** Horizontal scaling, stateless services
4. **Resilience:** Graceful degradation, circuit breakers, bulkheads
5. **Observability:** Full traceability of every transaction
6. **Domain-Driven Design:** Business logic encapsulated in bounded contexts
7. **API-First:** All functionality exposed through well-defined APIs
8. **Eventual Consistency:** Event-driven architecture for loose coupling

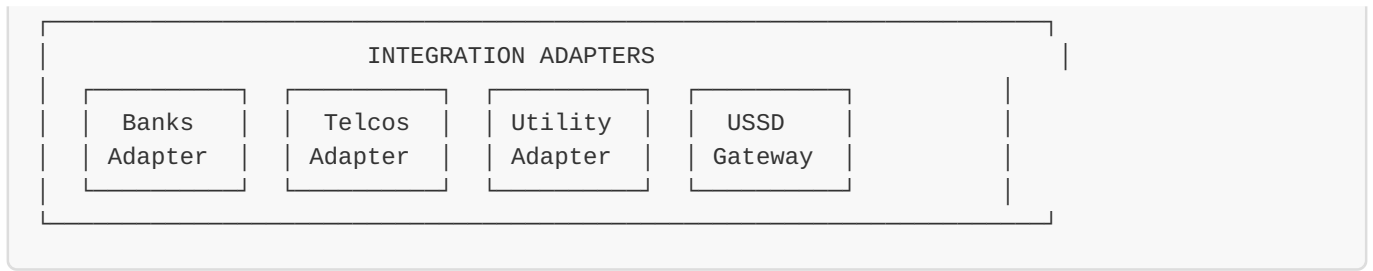
1.2 Technology Principles

- **Polyglot Persistence:** Right database for the right use case
 - **Infrastructure as Code:** All infrastructure versioned and automated
 - **Immutable Infrastructure:** Container-based deployments
 - **GitOps:** Declarative infrastructure and application deployment
 - **Automated Testing:** Unit, integration, contract, and chaos testing
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2. System Context

2.1 High-Level Architecture





2.2 Service Boundaries (Bounded Contexts)

1. Payment Processing Context

- P2P Transfer Service
- Cash-In/Out Service
- Merchant Payment Service

2. Account Management Context

- Wallet Service
- Customer Service
- KYC Service

3. Financial Context

- Ledger Service
- Fee Calculation Service
- Settlement Service

4. Risk Management Context

- Fraud Detection Service
- Limits Enforcement Service
- AML/Compliance Service

5. Notification Context

- Notification Service (SMS, Push, Email)
- Communication Gateway

6. Integration Context

- External Partner Adapters
- Channel Gateway Service

3. Architecture Patterns

3.1 CQRS (Command Query Responsibility Segregation)

Command Side (Write Model)

- Handles all state changes

- Strong consistency
- Domain-driven design
- Event sourcing optional

Query Side (Read Model)

- Optimized for reads
- Eventual consistency
- Denormalized views
- Multiple read models per service

Command → Aggregate → Event → Event Store → Projection → Read Model

3.2 Event Sourcing

Store all state changes as sequence of events:

```
-- Event Store Schema
CREATE TABLE EventStream (
  AggregateId UNIQUEIDENTIFIER NOT NULL,
  Version INT NOT NULL,
  EventType VARCHAR(100) NOT NULL,
  EventData NVARCHAR(MAX) NOT NULL,
  Metadata NVARCHAR(MAX),
  CreatedAt DATETIME2 NOT NULL DEFAULT GETUTCDATE(),
  PRIMARY KEY (AggregateId, Version)
);
```

Benefits:

- Complete audit trail
- Time travel debugging
- Event replay for new projections
- Temporal queries

3.3 Saga Pattern

For distributed transactions across services:

Orchestration-Based Saga:

P2P Transfer Saga:

1. Reserve sender balance
2. Create transfer record
3. Credit receiver wallet
4. Post ledger entries
5. Send notifications

Compensation Logic:

Each step has compensating transaction for rollback.

3.4 Outbox Pattern

Ensures reliable event publishing:

```
-- Same transaction writes business data + outbox
BEGIN TRANSACTION
  INSERT INTO Transactions (...) VALUES (...)
  INSERT INTO Outbox (EventType, Payload) VALUES (...)
COMMIT

-- Background worker polls and publishes
```

3.5 API Gateway Pattern

Single entry point with:

- Request routing
- Protocol translation
- Composition
- Authentication/Authorization
- Rate limiting
- Caching

3.6 Service Mesh Pattern

Infrastructure layer handling:

- mTLS between services
- Traffic management
- Observability
- Resilience (circuit breaker, retry)

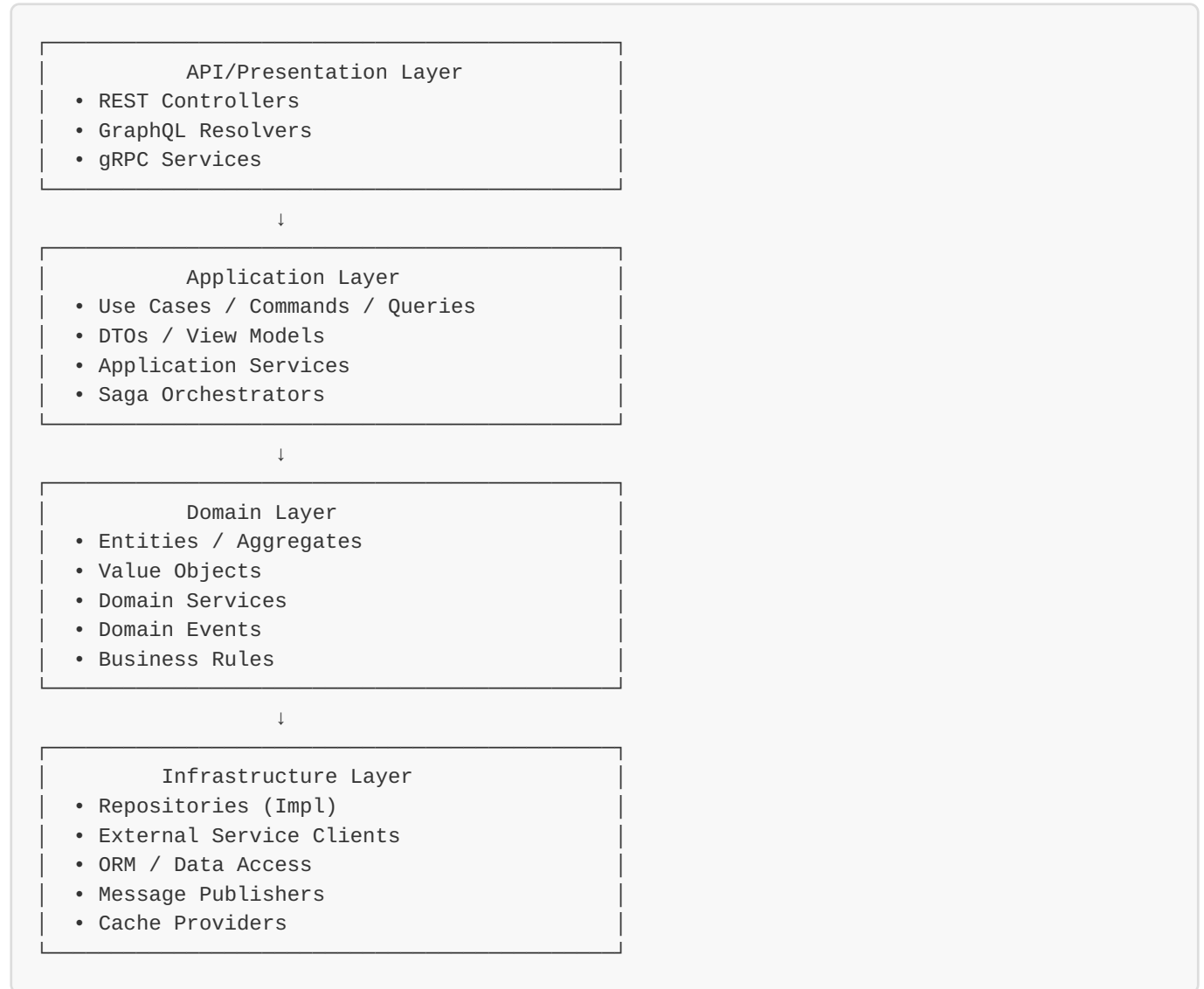
3.7 Strangler Fig Pattern

For gradual migration:

```
Legacy System → Proxy → Route to New/Old Service
```

4. Component Architecture

4.1 Service Structure (Clean Architecture)



4.2 P2P Transfer Service (Detailed)

Command Flow:

POST /api/v1/transfers/p2p

```

1. TransferController.InitiateP2P()
  ↓
2. TransferCommandHandler.Handle()
   - Validate request
   - Check idempotency
  ↓
3. TransferSaga.Execute()
   - Reserve sender balance
   - Verify receiver account
   - Apply limits
   - Check fraud rules
   - Calculate fees
  ↓
4. TransferAggregate.Execute()
   - Domain validation
   - Emit TransferInitiated event
   - Emit BalanceReserved event
  ↓
5. Repository.Save()
   - Write to event store
   - Write to outbox
   - Update read model
  ↓
6. Return 202 Accepted
   TransferId + Status URL

```

Event Handlers:

- `TransferCompleted` → Update wallet balances
- `TransferCompleted` → Post ledger entries
- `TransferCompleted` → Send notifications
- `TransferCompleted` → Update analytics

4.3 Ledger Service

Double-Entry Bookkeeping:

```

CREATE TABLE LedgerEntry (
    EntryId BIGINT IDENTITY PRIMARY KEY,
    TransactionId UNIQUEIDENTIFIER NOT NULL,
    AccountId UNIQUEIDENTIFIER NOT NULL,
    DebitAmount DECIMAL(18,2),
    CreditAmount DECIMAL(18,2),
    Balance DECIMAL(18,2) NOT NULL,
    EntryType VARCHAR(50) NOT NULL,
    Reference VARCHAR(200),
    CreatedAt DATETIME2 DEFAULT GETUTCDATE(),
    CONSTRAINT CHK_DebitOrCredit CHECK (
        (DebitAmount IS NOT NULL AND CreditAmount IS NULL) OR
        (DebitAmount IS NULL AND CreditAmount IS NOT NULL)
    )
);

-- For each transfer, create paired entries:
-- Entry 1: Debit sender account
-- Entry 2: Credit receiver account
-- Entry 3: Debit receiver for fee (optional)
-- Entry 4: Credit fee income account (optional)

```

Consistency:

- All entries in single transaction
- Running balance calculation
- Immutable ledger (no updates/deletes)

4.4 Wallet Service

Capabilities:

- Balance queries (cache-first)
- Account status management
- Balance reservations
- Multi-currency support

Balance Calculation:

```

public decimal GetBalance(Guid accountId) {
    // Try cache first
    var cached = _cache.Get($"balance:{accountId}");
    if (cached != null) return cached;

    // Fallback to ledger
    var balance = _ledgerService.ComputeBalance(accountId);

    // Cache with short TTL
    _cache.Set($"balance:{accountId}", balance, TimeSpan.FromSeconds(30));

    return balance;
}

```

4.5 Authentication & Authorization Service

OAuth 2.0 / OpenID Connect:

- Client credentials for service-to-service
- Authorization code for user apps
- JWT access tokens
- Refresh tokens

Token Structure:

```
{
  "sub": "customer:550e8400",
  "iss": "https://auth.nyotapay.com",
  "aud": ["api.nyotapay.com"],
  "exp": 1730000000,
  "iat": 1729996400,
  "scope": ["transfer:create", "wallet:read"],
  "roles": ["customer"],
  "kyc_level": "tier2"
}
```

RBAC + ABAC:

- Role-Based Access Control for coarse permissions
- Attribute-Based for fine-grained (e.g., KYC level)

4.6 Fraud Detection Service

Real-Time Fraud Checks:

1. Velocity checks (transaction frequency)
2. Amount anomaly detection
3. Geo-location analysis
4. Device fingerprinting
5. Behavioral biometrics
6. ML-based risk scoring

Decision Engine:

```
Risk Score < 30: Auto-approve
Risk Score 30-70: Manual review
Risk Score > 70: Auto-reject + Alert
```

4.7 Fee Calculation Engine

Dynamic Fee Structure:

```
public class FeeCalculator {
    public FeeResult Calculate(
        TransactionType type,
        decimal amount,
        CustomerTier tier,
        MerchantCategory merchant) {

        var rule = _ruleEngine.Match(type, tier, merchant);

        var fee = rule.Type switch {
            FeeType.Percentage => amount * rule.Rate,
            FeeType.Fixed => rule.FixedAmount,
            FeeType.Tiered => CalculateTiered(amount, rule.Tiers),
            _ => 0
        };

        return new FeeResult {
            FeeAmount = Math.Min(fee, rule.MaxFee),
            TaxAmount = fee * rule.TaxRate,
            NetFee = fee
        };
    }
}
```

5. Security Architecture

5.1 Defense in Depth

Layer 1: Network Security

- VPC / Network segmentation
- Firewall rules
- DDoS protection
- IDS/IPS

Layer 2: Edge Security

- WAF (ModSecurity/Cloudflare)
- Rate limiting
- IP allowlisting
- Bot protection

Layer 3: API Gateway

- Authentication (OAuth 2.0)
- Authorization (RBAC/ABAC)
- API key validation
- Request validation

Layer 4: Service Mesh

- mTLS between all services
- Service identity (SPIFFE)
- Fine-grained access control

Layer 5: Application

- Input validation
- SQL injection prevention
- XSS protection
- CSRF tokens

Layer 6: Data

- Encryption at rest (AES-256)
- Encryption in transit (TLS 1.3)
- Field-level encryption (PII)
- Key rotation

5.2 Zero Trust Architecture

Principles:

1. Never trust, always verify
2. Least privilege access
3. Microsegmentation
4. Continuous monitoring

Implementation:

Request → Identity Verification → Device Posture Check
→ Context Evaluation → Policy Decision → Allow/Deny

5.3 Secrets Management

HashiCorp Vault / Azure Key Vault:

Secrets Storage:

- Database credentials
- API keys
- Certificates
- Encryption keys

Features:

- Dynamic secrets (short-lived)
- Automatic rotation
- Audit logging
- Access policies

5.4 PCI DSS Compliance

Requirements Mapping:

- **Req 1-2:** Network segmentation, no default passwords
- **Req 3-4:** Encryption, secure transmission
- **Req 5-6:** Anti-malware, secure development
- **Req 7-9:** Access control, physical security
- **Req 10-12:** Logging, security policies

5.5 Cryptographic Standards

Encryption:

- Symmetric: AES-256-GCM
- Asymmetric: RSA-4096, ECC P-384
- Hashing: SHA-256, bcrypt (passwords)
- MAC: HMAC-SHA256

Key Management:

- Hardware Security Modules (HSM)
 - Key rotation every 90 days
 - Separate keys per environment
-

6. Data Architecture

6.1 Polyglot Persistence

SQL Server (Primary Transactional Data):

- Customer accounts
- Transactions
- Ledger entries
- KYC documents

Redis (Caching & Session):

- Session state
- Balance cache (30s TTL)
- Rate limit counters
- Distributed locks

Event Store (Event Sourcing):

- EventStoreDB or Kafka
- Immutable event log
- Aggregate event streams

Time-Series DB (Metrics):

- Prometheus
- Transaction metrics
- Performance data

Document Store (Optional - MongoDB):

- Audit logs
- Analytics data
- Configuration

Object Storage (S3/MinIO):

- KYC documents
- Transaction receipts
- Backups

6.2 Database Schema Design

Transaction Table:

```
CREATE TABLE [dbo].[Transaction] (
    TransactionId UNIQUEIDENTIFIER DEFAULT NEWID() PRIMARY KEY,
    ExternalRef VARCHAR(100) UNIQUE NOT NULL,
    TransactionType VARCHAR(50) NOT NULL,
    SourceAccountId UNIQUEIDENTIFIER NOT NULL,
    DestinationAccountId UNIQUEIDENTIFIER,
    Amount DECIMAL(18,2) NOT NULL,
    Currency CHAR(3) DEFAULT 'TZS',
    FeeAmount DECIMAL(18,2),
    Status VARCHAR(20) NOT NULL,
    FailureReason VARCHAR(500),
    Metadata NVARCHAR(MAX), -- JSON
    CreatedAt DATETIME2 DEFAULT GETUTCDATE(),
    UpdatedAt DATETIME2 DEFAULT GETUTCDATE(),
    CompletedAt DATETIME2,
    IdempotencyKey VARCHAR(100) UNIQUE,

    CONSTRAINT FK_Transaction_SourceAccount
        FOREIGN KEY (SourceAccountId)
        REFERENCES Account(AccountId),

    INDEX IX_Transaction_ExternalRef (ExternalRef),
    INDEX IX_Transaction_CreatedAt (CreatedAt),
    INDEX IX_Transaction_Status (Status)
);
```

Account Table:

```

CREATE TABLE [dbo].[Account] (
    AccountId UNIQUEIDENTIFIER DEFAULT NEWID() PRIMARY KEY,
    CustomerId UNIQUEIDENTIFIER NOT NULL,
    AccountNumber VARCHAR(20) UNIQUE NOT NULL,
    AccountType VARCHAR(20) NOT NULL,
    Currency CHAR(3) DEFAULT 'TZS',
    Status VARCHAR(20) NOT NULL,
    Balance AS (
        SELECT ISNULL(SUM(CreditAmount - DebitAmount), 0)
        FROM LedgerEntry
        WHERE AccountId = Account.AccountId
    ) PERSISTED,
    CreatedAt DATETIME2 DEFAULT GETUTCDATE(),

    CONSTRAINT FK_Account_Customer
    FOREIGN KEY (CustomerId)
    REFERENCES Customer(CustomerId)
);

```

Outbox Table:

```

CREATE TABLE [dbo].[Outbox] (
    OutboxId BIGINT IDENTITY PRIMARY KEY,
    AggregateId UNIQUEIDENTIFIER NOT NULL,
    EventType VARCHAR(100) NOT NULL,
    Payload NVARCHAR(MAX) NOT NULL,
    Status VARCHAR(20) DEFAULT 'Pending',
    RetryCount INT DEFAULT 0,
    CreatedAt DATETIME2 DEFAULT GETUTCDATE(),
    ProcessedAt DATETIME2,

    INDEX IX_Outbox_Status_CreatedAt (Status, CreatedAt)
);

```

6.3 Data Consistency Patterns

Strong Consistency (Synchronous):

- Wallet balance updates
- Ledger postings
- Transaction status

Eventual Consistency (Asynchronous):

- Notifications
- Analytics
- Reporting
- Search indexes

Saga Pattern for Distributed Transactions:

Transfer Saga:

1. Reserve sender balance (compensate: Unreserve)
2. Validate receiver (compensate: N/A)
3. Create transaction (compensate: Cancel transaction)
4. Post to ledger (compensate: Reverse entry)
5. Send notification (no compensation)

6.4 Data Retention & Archival

Hot Storage (SSD): Last 90 days

Warm Storage (HDD): 91 days - 7 years

Cold Storage (S3 Glacier): > 7 years

Regulatory Requirements:

- Transaction data: 7 years (PCI DSS)
- KYC documents: 5 years after account closure
- Audit logs: 1 year hot, 7 years cold

7. Integration Architecture

7.1 Integration Patterns

Adapter Pattern:

Core Service → Interface → Adapter → External System

Each adapter implements standard interface:

```
public interface IPaymentGateway {
    Task<PaymentResult> ProcessPayment(PaymentRequest request);
    Task<PaymentStatus> GetStatus(string referenceId);
    Task<RefundResult> RefundPayment(string referenceId);
}
```

7.2 Bank Integration

ISO 8583 Messaging:

MTI: 0200 (Financial Transaction Request)

Fields:

- 2: PAN
- 3: Processing Code
- 4: Amount
- 7: Transmission Date/Time
- 11: STAN
- 37: Reference Number
- ...

Async Settlement:

1. Real-time authorization
2. Batch settlement (EOD)
3. Reconciliation file exchange
4. Settlement confirmation

7.3 Telco Integration

USSD Gateway:

User → *123# → Telco USSD Gateway → NyotaPay USSD Service
← Menu Response ←

Mobile Money Integration:

- REST API (MTN, Airtel, Vodacom)
- OAuth 2.0 authentication
- Webhook callbacks for status

Airtime Topup:

Request → Adapter → Telco API → Instant Topup
Response ← Status ← Callback ←

7.4 Merchant Integration

Payment Gateway SDK:

```
const nyotaPay = new NyotaPay({ apiKey: 'xxx' });

const payment = await nyotaPay.createPayment({
  amount: 50000,
  currency: 'TZS',
  merchantRef: 'ORDER-12345',
  callbackUrl: 'https://merchant.com/callback'
});

// Redirect customer to payment.checkoutUrl
```

Webhook for Status:

```
POST /merchant/callback
{
  "eventType": "payment.completed",
  "paymentId": "550e8400-e29b-41d4-a716-446655440000",
  "merchantRef": "ORDER-12345",
  "amount": 50000,
  "status": "COMPLETED",
  "timestamp": "2025-11-05T10:30:00Z",
  "signature": "sha256=abc123..."
}
```

8. Deployment Architecture

8.1 Container Orchestration (Kubernetes)

Cluster Architecture:

```
Production Cluster:
├─ Master Nodes (3x) - Control Plane
│  ├─ API Server
│  ├─ Scheduler
│  └─ Controller Manager
│  └─ etcd cluster
├─ Worker Nodes (10x) - Application Workloads
│  ├─ Core Services (P2P, Cash, Merchant)
│  ├─ Shared Services (Wallet, Ledger, etc.)
│  └─ Integration Adapters
└─ Edge Nodes (3x) - Ingress
   ├─ Nginx Ingress Controller
   └─ API Gateway
```

Namespaces:

- **core** : Core business services
- **shared** : Shared capability services
- **integration** : External adapters
- **platform** : Infrastructure services (monitoring, logging)
- **security** : Security services

8.2 Service Deployment

Deployment Manifest Example:

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: p2p-service
  namespace: core
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  selector:
    matchLabels:
      app: p2p-service
  template:
    metadata:
      labels:
        app: p2p-service
        version: v1.2.0
    spec:
      containers:
        - name: p2p-service
          image: registry.nyotapay.com/p2p-service:1.2.0
          ports:
            - containerPort: 8080
              name: http
            - containerPort: 9090
              name: metrics
          env:
            - name: DATABASE_URL
              valueFrom:
                secretKeyRef:
                  name: db-credentials
                  key: connection-string
      resources:
        requests:
          cpu: 500m
          memory: 1Gi
        limits:
          cpu: 2000m
          memory: 4Gi
      livenessProbe:
        httpGet:
          path: /health/live
          port: 8080
        initialDelaySeconds: 30
        periodSeconds: 10
      readinessProbe:
        httpGet:
          path: /health/ready
          port: 8080
        initialDelaySeconds: 10
        periodSeconds: 5

```

8.3 Service Mesh (Istio)

Benefits:

- Automatic mTLS
- Traffic management
- Observability
- Policy enforcement

Virtual Service Example:

```
apiVersion: networking.istio.io/v1beta1
kind: VirtualService
metadata:
  name: p2p-service
spec:
  hosts:
  - p2p-service
  http:
  - match:
    - headers:
        x-api-version:
          exact: v2
    route:
    - destination:
        host: p2p-service
        subset: v2
        weight: 100
    - route:
        - destination:
            host: p2p-service
            subset: v1
            weight: 90
        - destination:
            host: p2p-service
            subset: v2
            weight: 10
  retries:
    attempts: 3
    perTryTimeout: 2s
    timeout: 10s
```

8.4 Infrastructure as Code

Terraform for Infrastructure:

```

resource "kubernetes_namespace" "core" {
  metadata {
    name = "core"
  }
}

resource "helm_release" "p2p_service" {
  name      = "p2p-service"
  namespace = kubernetes_namespace.core.metadata[0].name
  chart     = "../charts/p2p-service"

  values = [
    file("${path.module}/values/production.yaml")
  ]
}

```

GitOps with ArgoCD:

```

apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: nyotapay-core
spec:
  project: default
  source:
    repoURL: https://github.com/nyotapay/k8s-manifests
    targetRevision: HEAD
    path: production/core
  destination:
    server: https://kubernetes.default.svc
    namespace: core
  syncPolicy:
    automated:
      prune: true
      selfHeal: true

```

8.5 High Availability Setup

SQL Server:

AlwaysOn Availability Group:

- └─ Primary Replica (Node 1) - Read/Write
- └─ Secondary Replica (Node 2) - Async Sync + Readable
- └─ Secondary Replica (Node 3) - Async Sync + Readable

Quorum: Node + File Share Witness
 Listener: sql-nyotapay-ag.local:1433

Redis:

Redis Cluster (6 nodes):

- └─ Master 1 (Slots: 0-5460)
- └─ Master 2 (Slots: 5461-10922)
- └─ Master 3 (Slots: 10923-16383)
- └─ 3x Replicas (one per master)

Sentinel for automatic failover

Kafka:

Kafka Cluster (5 brokers):

- └─ Broker 1-3: Topic partitions
- └─ Broker 4-5: Additional replicas
- └─ Zookeeper Ensemble (3 nodes)

Replication Factor: 3

Min In-Sync Replicas: 2

9. Resilience & Scalability

9.1 Resilience Patterns

Circuit Breaker:

```
var circuitBreakerPolicy = Policy
    .Handle<HttpRequestException>()
    .CircuitBreakerAsync(
        exceptionsAllowedBeforeBreaking: 5,
        durationOfBreak: TimeSpan.FromSeconds(30),
        onBreak: (exception, duration) => {
            logger.LogWarning("Circuit breaker opened");
        },
        onReset: () => {
            logger.LogInformation("Circuit breaker reset");
        }
    );
```

Retry Policy:


```

var retryPolicy = Policy
    .Handle<HttpRequestException>()
    .WaitAndRetryAsync(
        retryCount: 3,
        sleepDurationProvider: attempt =>
            TimeSpan.FromSeconds(Math.Pow(2, attempt)),
        onRetry: (exception, timeSpan, retryCount, context) => {
            logger.LogWarning($"Retry {retryCount} after {timeSpan}");
        }
    );

```

Bulkhead Isolation:

```

var bulkheadPolicy = Policy
    .BulkheadAsync(
        maxParallelization: 10,
        maxQueuingActions: 20,
        onBulkheadRejectedAsync: async context => {
            await metricsCollector.IncrementBulkheadRejection();
        }
    );

```

Timeout:

```

var timeoutPolicy = Policy
    .TimeoutAsync(
        TimeSpan.FromSeconds(10),
        TimeoutStrategy.Pessimistic
    );

```

Combined Policy:

```

var resiliencePolicy = Policy.WrapAsync(
    circuitBreakerPolicy,
    retryPolicy,
    bulkheadPolicy,
    timeoutPolicy
);

await resiliencePolicy.ExecuteAsync(async () => {
    return await _httpClient.GetAsync(url);
});

```

9.2 Scalability Strategies

Horizontal Pod Autoscaling:

```

apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: p2p-service-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: p2p-service
  minReplicas: 3
  maxReplicas: 20
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 70
  - type: Resource
    resource:
      name: memory
      target:
        type: Utilization
        averageUtilization: 80
  - type: Pods
    pods:
      metric:
        name: http_requests_per_second
      target:
        type: AverageValue
        averageValue: "1000"

```

Vertical Pod Autoscaling:

```

apiVersion: autoscaling.k8s.io/v1
kind: VerticalPodAutoscaler
metadata:
  name: p2p-service-vpa
spec:
  targetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: p2p-service
  updatePolicy:
    updateMode: "Auto"
  resourcePolicy:
    containerPolicies:
    - containerName: p2p-service
      maxAllowed:
        cpu: 4
        memory: 8Gi
      minAllowed:
        cpu: 500m
        memory: 1Gi

```

Database Scaling:

- Read replicas for read-heavy queries
- Connection pooling (min: 10, max: 100)
- Query caching
- Database sharding (future)

Cache Scaling:

- Redis Cluster mode (horizontal scaling)
- Cache-aside pattern
- Write-through caching
- TTL-based invalidation

9.3 Load Balancing

Layer 7 Load Balancer (Nginx):

```

upstream p2p_service {
    least_conn;
    server p2p-1.core.svc.cluster.local:8080 weight=3;
    server p2p-2.core.svc.cluster.local:8080 weight=3;
    server p2p-3.core.svc.cluster.local:8080 weight=3;

    keepalive 32;
}

server {
    listen 443 ssl http2;
    server_name api.nyotapay.com;

    ssl_certificate /etc/ssl/certs/nyotapay.crt;
    ssl_certificate_key /etc/ssl/private/nyotapay.key;

    location /api/v1/transfers {
        proxy_pass http://p2p_service;
        proxy_http_version 1.1;
        proxy_set_header Connection "";
        proxy_set_header Host $host;
        proxy_set_header X-Request-ID $request_id;
    }
}

```

9.4 Rate Limiting

Distributed Rate Limiting (Redis):

```
public async Task<bool> AllowRequest(string clientId) {
    var key = $"rate_limit:{clientId}:{DateTime.UtcNow:yyyyMMddHHmm}";
    var count = await _redis.IncrementAsync(key);

    if (count == 1) {
        await _redis.ExpireAsync(key, TimeSpan.FromMinutes(1));
    }

    return count <= 100; // 100 requests per minute
}
```

Token Bucket Algorithm:

```
public class TokenBucketRateLimiter {
    private readonly int _capacity;
    private readonly int _refillRate;
    private int _tokens;
    private DateTime _lastRefill;

    public bool TryConsume(int tokens = 1) {
        Refill();

        if (_tokens >= tokens) {
            _tokens -= tokens;
            return true;
        }

        return false;
    }

    private void Refill() {
        var now = DateTime.UtcNow;
        var elapsed = (now - _lastRefill).TotalSeconds;
        var tokensToAdd = (int)(elapsed * _refillRate);

        _tokens = Math.Min(_capacity, _tokens + tokensToAdd);
        _lastRefill = now;
    }
}
```

9.5 Chaos Engineering

Chaos Mesh Experiments:

```

apiVersion: chaos-mesh.org/v1alpha1
kind: PodChaos
metadata:
  name: pod-failure-test
spec:
  action: pod-failure
  mode: one
  selector:
    namespaces:
      - core
    labelSelectors:
      app: p2p-service
  duration: "30s"
  scheduler:
    cron: "@every 1h"

```

Network Chaos:

```

apiVersion: chaos-mesh.org/v1alpha1
kind: NetworkChaos
metadata:
  name: network-delay-test
spec:
  action: delay
  mode: all
  selector:
    namespaces:
      - core
  delay:
    latency: "200ms"
    jitter: "50ms"
  duration: "5m"

```

10. API Design

10.1 RESTful API Standards

Base URL:

```
https://api.nyotapay.com/v1
```

Versioning:

- URI versioning: `/v1/` , `/v2/`
- Header versioning: `Accept: application/vnd.nyotapay.v1+json`

Resource Naming:

GET	/customers	# List customers
POST	/customers	# Create customer
GET	/customers/{id}	# Get customer
PUT	/customers/{id}	# Update customer
DELETE	/customers/{id}	# Delete customer
GET	/customers/{id}/accounts	# Nested resources
POST	/transfers/p2p	# Action resources
POST	/transfers/{id}/cancel	# Sub-actions

10.2 Request/Response Format**Standard Request:**

```
POST /api/v1/transfers/p2p
Content-Type: application/json
Authorization: Bearer eyJhbGc...
X-Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000
X-Request-ID: 7c9e6679-7425-40de-944b-e07fc1f90ae7

{
  "sourceAccountId": "550e8400-e29b-41d4-a716-446655440000",
  "destinationAccountId": "6ba7b810-9dad-11d1-80b4-00c04fd430c8",
  "amount": 50000,
  "currency": "TZS",
  "reference": "PAYMENT-12345",
  "description": "Payment for services",
  "metadata": {
    "channel": "mobile_app",
    "deviceId": "abc123"
  }
}
```

Success Response (202 Accepted):

```

HTTP/1.1 202 Accepted
Content-Type: application/json
X-Request-ID: 7c9e6679-7425-40de-944b-e07fc1f90ae7
Location: /api/v1/transfers/7c9e6679-7425-40de-944b-e07fc1f90ae7

{
  "transferId": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
  "status": "PROCESSING",
  "message": "Transfer is being processed",
  "statusUrl": "/api/v1/transfers/7c9e6679-7425-40de-944b-e07fc1f90ae7",
  "estimatedCompletion": "2025-11-05T10:35:00Z",
  "_links": {
    "self": { "href": "/api/v1/transfers/7c9e6679-7425-40de-944b-e07fc1f90ae7" },
    "cancel": { "href": "/api/v1/transfers/7c9e6679-7425-40de-944b-e07fc1f90ae7/cancel" }
  }
}

```

Error Response (400 Bad Request):

```

HTTP/1.1 400 Bad Request
Content-Type: application/problem+json
X-Request-ID: 7c9e6679-7425-40de-944b-e07fc1f90ae7

{
  "type": "https://api.nyotapay.com/errors/insufficient-balance",
  "title": "Insufficient Balance",
  "status": 400,
  "detail": "Account balance is insufficient for this transfer",
  "instance": "/api/v1/transfers/p2p",
  "traceId": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
  "errors": [
    {
      "field": "amount",
      "message": "Amount exceeds available balance of 30000 TZS"
    }
  ],
  "availableBalance": 30000,
  "requestedAmount": 50000
}

```

10.3 Idempotency

Idempotency Key Handling:

```

public async Task<IActionResult> CreateTransfer(
    [FromBody] TransferRequest request,
    [FromHeader(Name = "X-Idempotency-Key")] string idempotencyKey) {

    // Check if already processed
    var existing = await _repository.GetByIdempotencyKey(idempotencyKey);
    if (existing != null) {
        return StatusCode(existing.StatusCode, existing.Response);
    }

    // Process new request
    var result = await _transferService.Execute(request);

    // Store result with idempotency key
    await _repository.SaveIdempotencyResult(
        idempotencyKey,
        result,
        TimeSpan.FromHours(24));

    return Accepted(result);
}

```

10.4 Pagination

Cursor-Based Pagination:

```

GET /api/v1/transactions?limit=20&cursor=eyJpZCI6MTIzNDV9

Response:
{
  "data": [...],
  "pagination": {
    "nextCursor": "eyJpZCI6MTIzNjV9",
    "hasMore": true,
    "count": 20
  },
  "_links": {
    "next": { "href": "/api/v1/transactions?limit=20&cursor=eyJpZCI6MTIzNjV9" }
  }
}

```

10.5 Webhooks

Webhook Registration:


```
POST /api/v1/webhooks
{
  "url": "https://merchant.com/webhooks/nyotapay",
  "events": [
    "transfer.completed",
    "transfer.failed",
    "payment.completed"
  ],
  "secret": "whsec_abc123..."
}
```

Webhook Payload:

```
POST https://merchant.com/webhooks/nyotapay
Content-Type: application/json
X-NyotaPay-Signature: sha256=abc123...
X-NyotaPay-Event: transfer.completed

{
  "eventId": "evt_550e8400",
  "eventType": "transfer.completed",
  "timestamp": "2025-11-05T10:35:00Z",
  "data": {
    "transferId": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "status": "COMPLETED",
    "amount": 50000,
    "currency": "TZS"
  }
}
```

Signature Verification:

```
public bool VerifyWebhookSignature(
    string payload,
    string signature,
    string secret) {

    var hash = HMACSHA256.HashData(
        Encoding.UTF8.GetBytes(secret),
        Encoding.UTF8.GetBytes(payload)
    );

    var computedSignature = $"sha256={Convert.ToHexString(hash)}";
    return signature.Equals(computedSignature, StringComparison.OrdinalIgnoreCase);
}
```

11. Observability

11.1 Three Pillars

Metrics → Logs → Traces

11.2 Metrics (Prometheus + Grafana)

Application Metrics:

```
// Counter
private static readonly Counter TransferRequests = Metrics
    .CreateCounter("nyotapay_transfer_requests_total",
        "Total transfer requests",
        new CounterConfiguration {
            LabelNames = new[] { "type", "status" }
        });

// Histogram
private static readonly Histogram TransferDuration = Metrics
    .CreateHistogram("nyotapay_transfer_duration_seconds",
        "Transfer processing duration",
        new HistogramConfiguration {
            LabelNames = new[] { "type" },
            Buckets = new[] { 0.1, 0.5, 1, 2, 5, 10 }
        });

// Gauge
private static readonly Gauge ActiveTransfers = Metrics
    .CreateGauge("nyotapay_active_transfers",
        "Number of active transfers");

// Usage
TransferRequests.WithLabels("p2p", "success").Inc();
using (TransferDuration.WithLabels("p2p").NewTimer()) {
    await ProcessTransfer();
}
```

Infrastructure Metrics:

- CPU, Memory, Disk usage
- Network I/O
- Pod restarts
- Request rate, error rate, duration (RED)

Business Metrics:

- Transaction volume
- Revenue (by type, by merchant)
- Customer acquisition
- Churn rate

Grafana Dashboards:

- Service overview (SLIs, error rate, latency)
- Transaction monitoring
- Infrastructure health
- Business KPIs

11.3 Logging (ELK Stack / Loki)

Structured Logging:

```

_logger.LogInformation(
    "Transfer initiated. " +
    "TransferId={TransferId}, " +
    "SourceAccount={SourceAccount}, " +
    "Amount={Amount}, " +
    "RequestId={RequestId}",
    transferId,
    sourceAccount,
    amount,
    requestId
);

// Output (JSON):
{
  "@timestamp": "2025-11-05T10:30:00.123Z",
  "level": "Information",
  "message": "Transfer initiated",
  "transferId": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
  "sourceAccount": "550e8400-e29b-41d4-a716-446655440000",
  "amount": 50000,
  "requestId": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
  "service": "p2p-service",
  "version": "1.2.0",
  "environment": "production"
}

```

Log Levels:

- **TRACE**: Very detailed (dev only)
- **DEBUG**: Diagnostic information (dev/staging)
- **INFO**: General operational events
- **WARN**: Recoverable issues
- **ERROR**: Unhandled errors
- **FATAL**: Critical failures

Centralized Logging:

```

Application → Fluent Bit → Elasticsearch → Kibana
              (sidecar)   (index)           (visualize)

```

11.4 Distributed Tracing (Jaeger/Zipkin)

Trace Context Propagation:

```
using var activity = Activity.Current?.Source.StartActivity("ProcessTransfer");
activity?.SetTag("transfer.id", transferId);
activity?.SetTag("transfer.type", "p2p");
activity?.SetTag("transfer.amount", amount);

try {
    var result = await _walletService.ReserveBalance(sourceAccount, amount);
    activity?.SetTag("balance.reserved", true);

    // ... continue processing

    activity?.SetStatus(ActivityStatusCode.Ok);
    return result;
}
catch (Exception ex) {
    activity?.SetStatus(ActivityStatusCode.Error, ex.Message);
    activity?.RecordException(ex);
    throw;
}
```

Trace Visualization:

```
[API Gateway] → [P2P Service] → [Wallet Service] → [Ledger Service]
    |           |           |           |
    10ms        25ms        15ms        30ms
                                     |
                                [SQL Server]
                                    50ms

Total: 130ms
```

Key Traces:

- Request-to-response flow
- Cross-service calls
- Database queries
- External API calls
- Error paths

11.5 Alerting

Prometheus Alert Rules:

```

groups:
- name: nyotapay_alerts
  interval: 30s
  rules:
- alert: HighErrorRate
  expr: |
    rate(nyotapay_transfer_requests_total{status="error"}[5m]) > 0.05
  for: 2m
  labels:
    severity: critical
  annotations:
    summary: "High error rate detected"
    description: "Error rate is {{ $value | humanizePercentage }} for {{ $labels.type }}"

- alert: SlowTransferProcessing
  expr: |
    histogram_quantile(0.95,
      rate(nyotapay_transfer_duration_seconds_bucket[5m])
    ) > 5
  for: 5m
  labels:
    severity: warning
  annotations:
    summary: "Slow transfer processing"
    description: "P95 latency is {{ $value }}s"

- alert: ServiceDown
  expr: up{job="p2p-service"} == 0
  for: 1m
  labels:
    severity: critical
  annotations:
    summary: "Service is down"

```

Alert Channels:

- PagerDuty (critical, on-call)
- Slack (warning, info)
- Email (daily digest)
- SMS (critical only)

11.6 SLIs & SLOs

Service Level Indicators:

- **Availability:** % of successful requests
- **Latency:** P95 response time < 500ms
- **Error Rate:** < 0.1% of requests fail
- **Throughput:** Requests per second

Service Level Objectives:

P2P Transfer Service:

- Availability: 99.95% (21.6 minutes downtime/month)
- Latency (P95): < 500ms
- Latency (P99): < 1000ms
- Error Rate: < 0.05%

Wallet Service:

- Availability: 99.99% (4.32 minutes downtime/month)
- Latency (P95): < 100ms
- Error Rate: < 0.01%

Error Budget:

Monthly Error Budget = $(1 - \text{SLO}) \times \text{Total Requests}$

Example:

SLO = 99.95%

Total Requests = 10M

Error Budget = $0.0005 \times 10\text{M} = 5,000$ failed requests

12. Disaster Recovery

12.1 Backup Strategy

Database Backups:

- **Full Backup:** Daily at 2 AM
- **Differential Backup:** Every 6 hours
- **Transaction Log Backup:** Every 15 minutes
- **Retention:** 7 days local, 30 days offsite, 7 years archived

Event Store Backups:

- **Snapshot:** Daily
- **Event Log:** Continuous replication
- **Retention:** Indefinite (legal requirement)

Configuration Backups:

- Infrastructure as Code (Git)
- Secrets (encrypted backups)
- Kubernetes manifests (Git)

12.2 Recovery Procedures

RTO (Recovery Time Objective):

- Critical services: 15 minutes

- Non-critical services: 1 hour
- Full system: 4 hours

RPO (Recovery Point Objective):

- Transactional data: 1 minute (transaction log backup)
- Configuration: Real-time (Git)

Disaster Scenarios:

1. Database Failure:

1. Failover to secondary replica (automatic, <30s)
2. Verify data consistency
3. Investigate and fix primary
4. Failback when stable

2. Availability Zone Failure:

1. Kubernetes reschedules pods to healthy nodes (2-5 min)
2. Traffic rerouted to available zones
3. Monitor and scale if needed

3. Complete Data Center Failure:

1. Activate DR site
 2. Restore latest backups
 3. Replay transaction logs
 4. Update DNS to DR site
 5. Resume operations
- Estimated: 2-4 hours

12.3 Business Continuity

Cold Standby:

- Infrastructure provisioned but not running
- Database backups replicated
- Manual activation process

Warm Standby:

- Minimal infrastructure running
- Database replication active
- Quick scale-up capability

Hot Standby (Recommended):

- Full infrastructure running
- Active-active database replication

- Automatic failover
- Geo-distributed deployment

12.4 Testing

DR Drills:

- Quarterly full DR test
- Monthly failover test
- Weekly backup restore test

Chaos Engineering:

- Pod failure injection
 - Network latency/partition
 - Database connection loss
 - Resource exhaustion
-

Appendix

A. Technology Stack

Backend Services:

- .NET 8 / ASP.NET Core
- C# 12

Databases:

- SQL Server 2022 (primary)
- Redis 7 (cache)
- EventStoreDB (event sourcing)

Messaging:

- Apache Kafka / RabbitMQ

API Gateway:

- Kong / Apigee

Service Mesh:

- Istio / Linkerd

Container Orchestration:

- Kubernetes 1.28+
- Helm 3

Observability:

- Prometheus + Grafana

- ELK Stack / Loki
- Jaeger / Tempo

CI/CD:

- GitLab CI / GitHub Actions
- ArgoCD (GitOps)
- Terraform (IaC)

Security:

- HashiCorp Vault
- Cert-Manager
- OAuth 2.0 / OpenID Connect

B. Glossary

- **Aggregate:** DDD pattern representing a cluster of objects treated as a single unit
- **CQRS:** Command Query Responsibility Segregation
- **Event Sourcing:** Storing state changes as events
- **Idempotency:** Same operation produces same result regardless of repetition
- **Saga:** Pattern for managing distributed transactions
- **Service Mesh:** Infrastructure layer for service-to-service communication
- **SLI/SLO/SLA:** Service Level Indicator/Objective/Agreement

C. References

1. Microsoft Azure Architecture Center
 2. AWS Well-Architected Framework
 3. Google Cloud Architecture Framework
 4. Domain-Driven Design by Eric Evans
 5. Building Microservices by Sam Newman
 6. Designing Data-Intensive Applications by Martin Kleppmann
 7. Site Reliability Engineering (SRE) by Google
-

Document Control

Version	Date	Author	Changes
1.0	2024-10-01	Architecture Team	Initial version
2.0	2025-11-05	Architecture Team	Complete redesign with DDD, CQRS, Event Sourcing

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